Supplementary Material
Does Explicit Monocular 3D Geometry Help AI Generated Image Detectors?

S1 Split summary and leakage report

Table S1 is produced from `split_summary.csv`; the leakage checks come from `leakage_report.json`. The run stops with an error if any check fails, so a completed run implies that all overlaps below are empty.

Table S1: Images per set, class and generator or real source after capping and deduplication.

Set	Generator or real source	Images
Fit	[xyz]	[xyz]
Validation	[xyz]	[xyz]
Test A	[xyz]	[xyz]
Test B	[xyz]	[xyz]
Test C	[xyz]	[xyz]
Test D	[xyz]	[xyz]

Table S2: Leakage report.

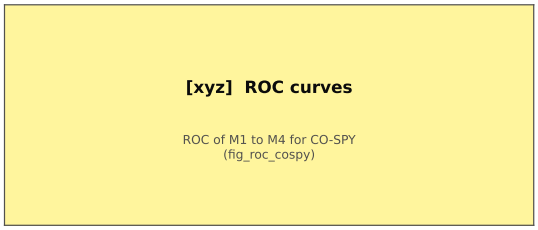
Check	Value
Generators shared by fit pool and tests	[xyz] (must be none)
Real image sources shared by fit pool and tests	[xyz] (must be none)
Captions shared by fit pool and tests	[xyz] (must be 0)
Captions shared by fit and validation	[xyz] (must be 0)
Fit images removed as near duplicates of test images	[xyz]
Images that could not be read and were dropped	[xyz]

S2 Normal convention and estimator checks

The sign vector aligning DSINE normals with MoGe normals was [xyz], with a median angle of [xyz] degrees on 50 fit images (`geometry_convention.json`). Missing value flag rates per class: G2 [xyz] (real) and [xyz] (generated), G3 [xyz] and [xyz], G4 [xyz] and [xyz].

Table S3: Selected regularisation C, generator grouped cross validation AUROC, temperature and threshold for each base detector and model (from `stackers/*/*.json`).

Detector	Model	C	CV AUROC	Temperature	Threshold (logit)
B-Free	M1 to M5	[xyz]	[xyz]	[xyz]	[xyz]
CO-SPY	M1 to M5	[xyz]	[xyz]	[xyz]	[xyz]
UnivFD	M1 to M5	[xyz]	[xyz]	[xyz]	[xyz]



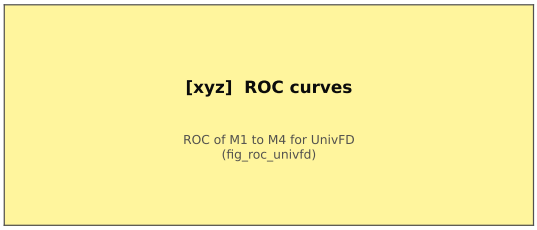


Figure S1: ROC curves of M1 to M4 for CO-SPY (left) and UnivFD (right), unseen generators.

S3 Stacker selection

S4 Additional results

S5 Supplementary image level tests

DeLong tests for M3 against M1 on pooled unseen generators gave z = [xyz] (p = [xyz]) for B-Free, [xyz] ([xyz]) for CO-SPY and [xyz] ([xyz]) for UnivFD (`hypotheses.json`, key `delong_M3_vs_M1`). These image level tests ignore the clustering of images within generators and are reported only as a secondary analysis.

S6 Outside reference: zero shot multimodal model

Qwen2.5-VL-7B-Instruct (Bai et al., 2025), asked whether each image is AI generated and scored by the log probability ratio of the answers Yes and No, reached a pooled AUROC of [xyz] on [xyz] sampled test images. It is not part of any hypothesis test.

References

Shuai Bai et al. Qwen2.5-VL Technical Report. arXiv preprint arXiv:2502.13923, 2025. URL `https://arxiv.org/abs/2502.13923`.

Table S4: Seen family generators (excluded from the headline analysis): macro AUROC of M1 and M3.

Detector	M1	M3
B-Free	[xyz]	[xyz]
CO-SPY	[xyz]	[xyz]
UnivFD	[xyz]	[xyz]

Table S5: Calibration (ECE, 15 bins) and accuracy on real and generated images at the fixed threshold, clean copies, pooled unseen generators.

Detector	Model	ECE	Real accuracy	Generated accuracy
B-Free	M1 / M3	[xyz] / [xyz]	[xyz] / [xyz]	[xyz] / [xyz]
CO-SPY	M1 / M3	[xyz] / [xyz]	[xyz] / [xyz]	[xyz] / [xyz]
UnivFD	M1 / M3	[xyz] / [xyz]	[xyz] / [xyz]	[xyz] / [xyz]